POINTWISE CHARACTERIZATIONS OF GEOMETRIC REGULARITY BY PURELY INSEPARABLE BASE CHANGE AND BY THE LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and closes the cycle of equivalences; no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-regular-at-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Geometric regularity of a k-algebra. Algebra, Definition algebra-definition-geometrically-regular, invoked at proof lines 2409-2410 and named in the statement. Supplies the algebraic property identified with geometric regularity of the local ring.
- Spectrum of a purely inseparable base change and localness of the tensor product. Algebra, Lemmas algebra-lemma-p-ring-map and algebra-lemma-characterize-local-ring, invoked at proof lines 2403-2406. Supplies that the tensor product of the local ring with a finite purely inseparable extension is again local, with the same spectrum.
- Surjectivity is stable under base change. Morphisms, Lemma morphisms-lemma-base-change-surjective, invoked at proof lines 2413-2416. Supplies the existence of a point above x after an arbitrary field extension.

Source reference. Original definition 0382: sources/stacks-chow/tags.txt, source lines 4221–4221. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00E9: sources/stacks-chow/tags.txt, source lines 526–526. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally Noetherian scheme over k. Let $x \in X$. The following are equivalent*

(1) *X is geometrically regular at x,*
(2) *for every finite purely inseparable field extension k' of k and $x' \in X_{k'}$ lying over x the local ring $\mathcal{O}_{X_{k'},x'}$ is regular, and*
(3) *the ring $\mathcal{O}_{X,x}$ is geometrically regular over k (see Algebra, Definition 0382).*

Proof. It is clear that (1) implies (2). Assume (2). This in particular implies that $\mathcal{O}_{X,x}$ is a regular local ring. Let k'/k be a finite purely inseparable field extension. Consider the ring $\mathcal{O}_{X,x} \otimes_k k'$. By Algebra, Lemma 0BRA its spectrum is the same as the spectrum of $\mathcal{O}_{X,x}$. Hence it is a local ring also (Algebra, Lemma 00E9). Therefore there is a unique point $x' \in X_{k'}$ lying over x and $\mathcal{O}_{X_{k'},x'} \cong \mathcal{O}_{X,x} \otimes_k k'$. By assumption this is a regular ring. Hence we deduce (3) from the definition of a geometrically regular ring.

Assume (3). Let k'/k be a field extension. Since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is surjective, also $X_{k'} \to X$ is surjective (Morphisms, Lemma 01S1). Let $x' \in X_{k'}$ be any point lying over x. The local ring $\mathcal{O}_{X_{k'},x'}$ is a localization of the ring $\mathcal{O}_{X,x} \otimes_k k'$. Hence it is regular by assumption and (1) is proved. □